

Mesenchymal stem cells derived extracellular vesicles: A promising nanomedicine for drug delivery system

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Abstract

An ideal drug delivery system should selectively deliver incorporated therapeutics to the target site, escape from immune cells recognition and degradation, and act controlled release of incorporated therapeutics in the site targeted. Extracellular vesicles (EVs) have gained great attention for their potential application as a drug delivery system in nanomedicine. EVs such as exosomes are membrane-bound vesicles that contribute to intracellular communication by transferring various biomolecules including RNAs, proteins, and lipids. EVs derived from mesenchymal stem cells (MSCs-EVs) have several advantages such as low immunogenicity, high biocompatibility, and stability against conventional synthetic carriers, opening new avenues for delivering therapeutic agents to target cells. To obtain modified MSCs-EVs, several loading methods are used to incorporate different therapeutic agents including proteins, RNAs, and chemotherapeutic drugs into MSCs-EVs. In addition, modification of MSCs-EVs surface may improve their potential in targeted therapies. Modified MSCs-EVs have been shown to improve many diseases including, cancer, cardiovascular diseases, and diabetes mellitus. While land greatly potential, the application of MSCs-EVs as a drug-delivery system has been hampered by several challenges. Clinical translation of modified-EVs needs further scrutiny. In this review, we discuss the biogenesis and production of EVs along with the loading and modification methods of MSCs-EVs. We also describe numerous MSCs-EVs based delivery studies with a focus on advantages and challenges when using them as a drug delivery system.